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Investigation of mechanical behavior and damage process of 2D C/SiC composites under cyclic loading: Experiment and simulation

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ABSTRACT

2D woven C/SiC composites exhibit nonlinear mechanical behavior under cyclic loading, accompanied by complex damage processes and failure modes. In this study, the mechanical behavior and damage process of 2D C/SiC composites under cyclic loading were investigated by experimental and numerical methods. First, loading-unloading experiments were conducted. The damage evolution of the composites was monitored using acoustic emission (AE) and digital image correlation (DIC), and the damage modes were identified through *k*-means cluster analysis: matrix cracking [0–180 kHz], fiber pull-out [180–290 kHz], interface debonding [290–390 kHz], and fiber breakage [430–900 kHz]. Then, an anisotropic elastic-plastic damage constitutive model was proposed and implemented with an implicit solution method via a user-defined subroutine (UMAT) in ABAQUS. The stress-strain curve of the composites under cyclic loading was analyzed, and the evolution processes of residual strain and unloading modulus were characterized. Additionally, the damage process and failure modes of the composites were examined based on simulation results. The predicted results from the model showed good agreement with experimental observations.

1. Introduction

Ceramic Matrix Composites (CMCs) have garnered significant attention in aerospace and other advanced engineering fields due to their superior mechanical strength, high-temperature capability, ablation resistance, and low density. 2D woven C/SiC composites are widely used for thermal protection structures of reusable aerospace vehicles [1–3]. Under cyclic loading, internal damage accumulates continuously, inducing sophisticated evolution of residual mechanical properties. For example, prolonged cyclic loading leads to residual deformation increasing and unloading modulus decreasing. However, residual strength has more complex evolution process, which typically increases first and then decreases [4,5]. These complex damage mechanisms should be in-depth investigated. Therefore, exploring the reusable capability of CMCs and clarifying their intrinsic damage evolution mechanisms are challenging yet of great theoretical and engineering significance.

Many studies have employed advanced experimental techniques and nondestructive testing methods, such as radiography [6,7], infrared thermography [8], and acoustic emission (AE) [9], to elucidate the nonlinear responses and damage evolution of composites [10]. AE, in particular, allows real-time detection of internal damage without requiring external excitation. Research has shown that damage sources in the composites release energy, which is associated with the emission of transient elastic waves observable via AE. These waves can be used to identify different failure modes [10]. There have been wide application cases in the failure analysis of CMCs with AE method under typical loading conditions, such as tension [11–13], compression [14–17],shear [18,19], and bending [20–22]. However, studies on the evolution of various damage modes under cyclic loading are still insufficient. Therefore, integrating AE monitoring with advanced clustering algorithms is essential for uncovering the synergistic relationship between macroscopic mechanical responses and microscopic damage mechanisms during cyclic loading, and providing a reliable experimental foundation for understanding the failure mechanisms of CMCs under cyclic loading.

Other scholars have developed theoretical models or numerical methods to study the nonlinear response and damage behavior of CMCs

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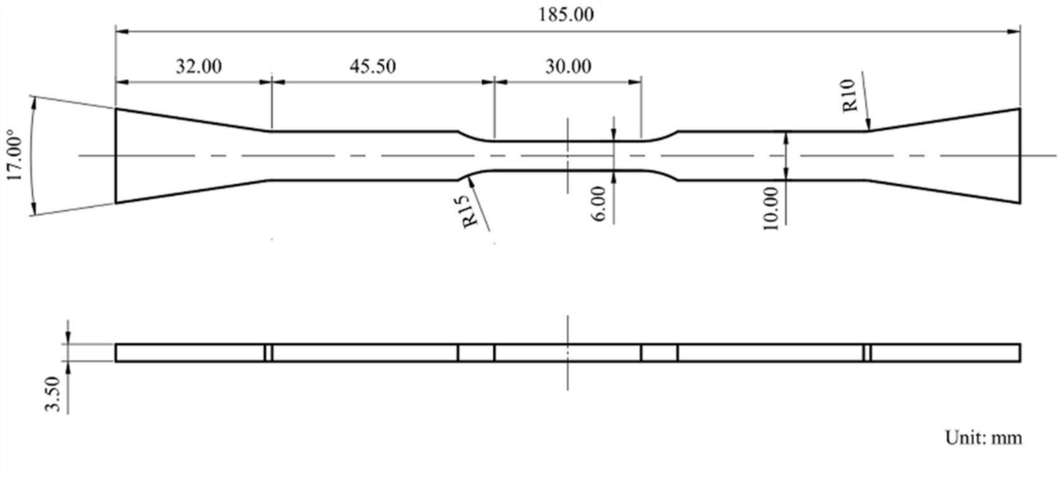

Fig. 1. Geometry of the tensile specimen.

under cyclic loading. Many theoretical models [23–25] have been developed, based on the hysteresis-based approach to estimate the mechanical response of CMCs. However, although these models effectively capture stress-strain behavior, they provide limited insight into the internal damage processes. Numerical methods, by contrast, present complementary advantages and enable intuitive visualization of damage initiation and propagation. Wang et al. [26] considered different failure modes and developed a multiscale progressive damage model for 2D woven C/SiC composites, predicting the fatigue life of 2D woven C/SiC composites under random loading. Yang et al. [27] established a damage constitutive model in macroscale and investigated the damage evolution law, strain response and rupture strength under incremental cyclic tension along both on-axis and off-axis directions. Shen et al. [28] employed a multiscale approach to investigate the impact of pore defects on the overall performance of C/SiC composites. Li et al. [29] introduced a matrix cracking strain-saturation mechanism to describe the longitudinal modulus degradation of fiber bundles and simulate the tensile failure behavior of CMCs. However, most existing works often overlook the complex evolution of residual performance under cyclic loading conditions. In particular, the evolution process of residual strain, stiffness degradation, and internal damage mechanisms during cyclic loading condition, remain insufficiently explored. These phenomena are critical to understanding the long-term performance and reusable capability of C/SiC composites.

In the present study, the cyclic mechanical behavior of 2D woven C/SiC composites fabricated via the polymer infiltration and pyrolysis (PIP) process is investigated. Loading-unloading experiments were conducted, and full-field strain evolution was captured using DIC method. AE monitoring and fracture-surface analyses were employed to identify damage modes. Based on the failure modes obtained in experiments, an anisotropic elastoplastic damage constitutive model was developed with an implicit solution method to simulate the nonlinear response under cyclic loading. A mesoscale representative volume cell (RVC) was established to predict the evolution of residual strain and unloading modulus degradation. Additionally, the damage process and failure modes of the composites were predicted and compared with the results of experiments.

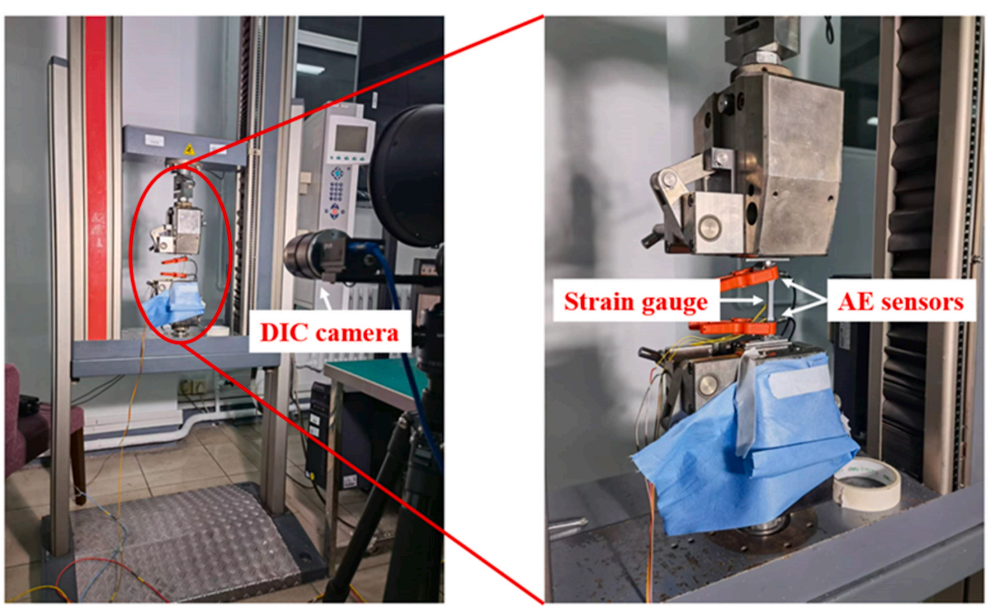

Fig. 2. Experiment devices.

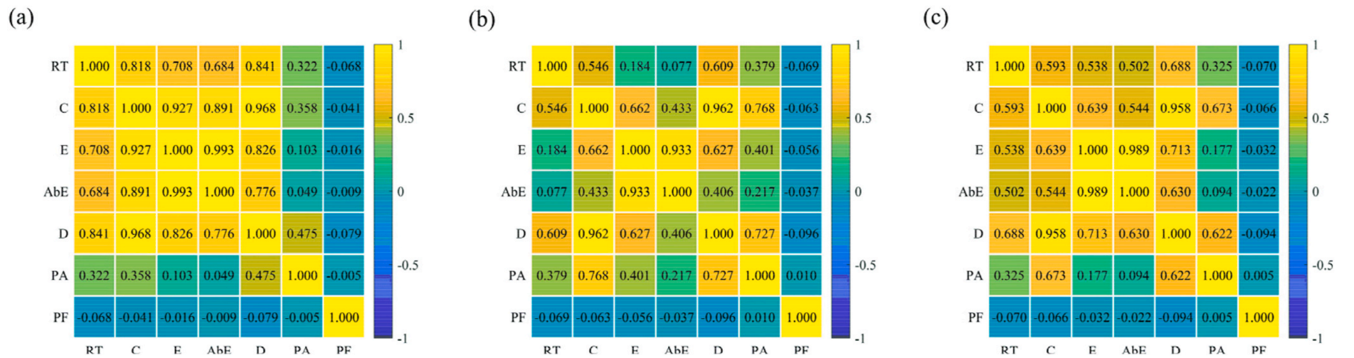

Fig. 3. Pearson correlation coefficients between AE parameters: (a) sample 1, (b) sample 2, (c) sample 3.

2. Materials and experiments

2.1. Materials and specimens

The 2D woven C/SiC composites used in this investigation were manufactured at the Shanghai Institute of Ceramics, Chinese Academy of Sciences. The composites were prepared using the polymer infiltration and pyrolysis process, with a fiber volume fraction of approximately 40 %. The specimens were cut into dog-bone shapes, as shown in Fig. 1. To prevent surface damage during tensile tests, the edges of the specimens were protected by bonding two pairs of aluminum tabs.

2.2. Experimental procedure

Experiments were conducted at room temperature using a Zwick/Roell Z100 universal mechanical testing machine. The tests were conducted in accordance with ASTM C1359-13. Loading was controlled using crosshead displacement at a constant speed of 0.5 mm/min. Full-field strain evolution was recorded using DIC, with strain gauges placed at the center of the specimens for comparison with DIC results. The AE technique was employed to monitor the tests in real-time. The experimental device was shown in Fig. 2. DIC and VIC-3D software (Correlated Solutions Inc., USA) were used to process the strain data. A black speckled pattern, applied using an airbrush, ensured randomness and uniformity across the specimen surface. Two CCD digital cameras (FLIR Inc., USA) with a resolution of 2048 × 2048 pixels^2 were used to capture images, which were then processed using the VIC-3D software to obtain displacement and strain fields. AE sensors from Physical Acoustics Corporation (PAC) were used to detect internal acoustic signals during the tests. Silicon high vacuum grease was applied as a coupling agent at the sensor/specimen interface. According to ASTM E976 standards, coupling verification was conducted using a lead-break test. The AE signal threshold was set at 40 dB to filter background noise. The AE software calculated signal descriptors such as peak amplitude, energy, count, rise time, and duration.

Before conducting the incremental loading/unloading tests, uniaxial tensile tests were first performed. 5 samples were tested. The results of uniaxial tensile are shown in Appendix A. These results indicated that the C/SiC composites were highly consistent in their tensile behavior, exhibiting minimal variability. Then, cyclic loading-unloading tests were performed with an incremental step loading, up to final rupture. A total of five specimens were tested, and three valid test results were obtained. The results of three valid tests showed good consistency. As for the two sets of invalid data, their failure strengths were significantly lower than expected. The primary reason for the low failure strength in these two sets of data was due to the specimen misalignment. The results of incremental loading/unloading tests can be found in Appendix A.

2.3. AE signal analysis procedure

2.3.1. Feature component selection

Feature components were selected to reduce the dimensionality of the data, which was crucial for clustering accuracy. AE time-domain parameters, such as rise time (RT), count (C), energy (E), absolute energy (AbE), duration (D) and peak amplitude (PA), along with frequency-domain parameters obtained through Fast Fourier Transform (FFT), such as peak frequency (PF), were calculated by the AE software. Pearson correlation coefficient method was used to analyze the correlation between these parameters [30].

The Pearson correlation coefficient ρ_{xy} quantifies the linear correlation between two variables and is calculated as follows:

$$\rho_{xy} = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \overline{y})^2}} \tag{1}$$

where x_i and y_i are the variables, and $\overline{x}, \overline{y}$ are their respective means. The Pearson coefficient ranges from −1 to 1. A value close to 1 indicates a strong positive correlation, while a value close to −1 indicates a strong negative correlation. If $\rho_{xy} = 0$, there is no linear correlation between the variables.

The Pearson correlation coefficients between AE parameters were calculated, as shown in Fig. 3. Among all AE parameters, PF and PA exhibited negligible linear correlation with the remaining variables, indicating their relative independence. Therefore, PF and PA were selected as feature components for subsequent analysis.

2.3.2. Determination of cluster number

To determine the optimal number of clusters, which significantly impacts the accuracy of clustering results, the Calinski-Harabasz index (CH index) and Davies-Bouldin Index (DB index) were used.

The CH index evaluates clustering performance by considering both the compactness of clusters and their separation. It is calculated as the ratio of the between-cluster variance to within-cluster variance:

$$CH(k) = \frac{tr(\boldsymbol{B}_k)}{tr(\boldsymbol{W}_k)} \cdot \frac{N_{total} - k}{k - 1} \tag{2}$$

where N_{total} is the total number of data points, k is the number of clusters, and $\mathrm{tr}(\boldsymbol{B}_k)$ and $\mathrm{tr}(\boldsymbol{W}_k)$ represent the between-cluster and within-cluster variance matrices, respectively. A higher CH index suggests better-defined and separated clusters. The DB index assesses clustering by comparing the average similarity within clusters to the maximum similarity between clusters:

$$DB(k) = \frac{1}{k}\sum_{i=1}^{k}\max_{j \neq i}(R_{ij}) \tag{3}$$

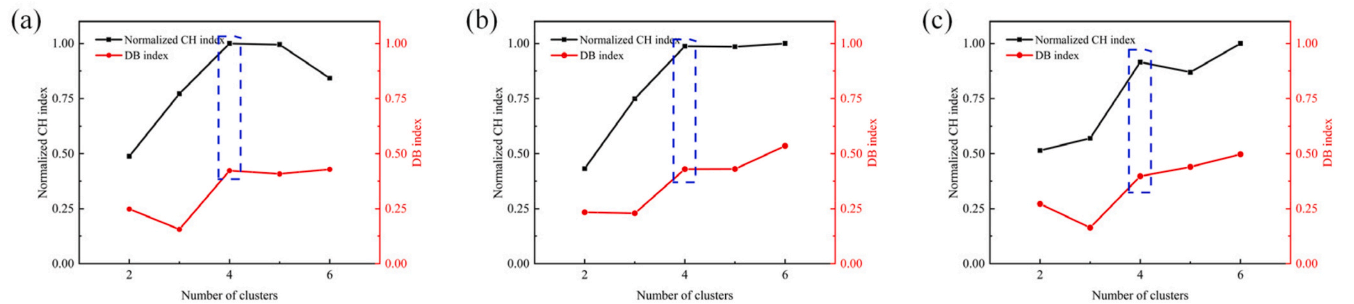

Fig. 4. Number of clusters evaluated by CH index and DB index: (a) sample 1, (b) sample 2, and (c) sample 3.

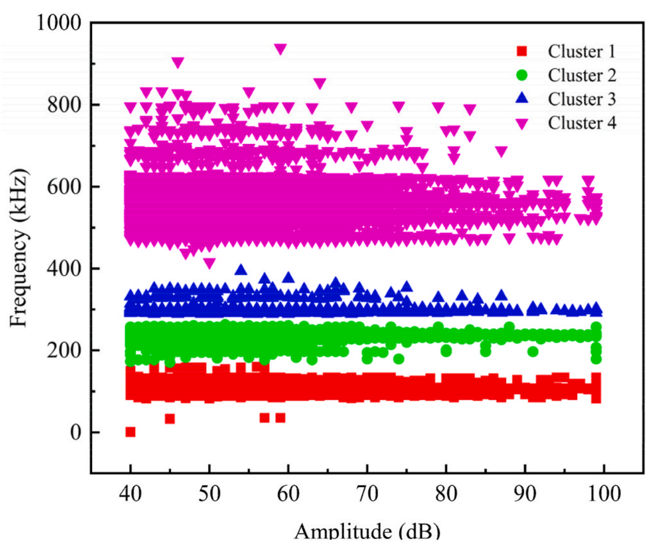

Fig. 5. Cluster result of the typical specimen by *k*-means method.

where R_{ij} is the similarity between clusters i and j. A lower DB index indicates more well-separated clusters. The CH index and DB index were calculated for all AE signals of the specimens, as shown in Fig. 4. Based on the results, four clusters were identified as the optimal number. These four clusters could effectively capture the primary damage modes, observed in the C/SiC composites during incremental loading/unloading.

2.3.3. Cluster analysis

The *k*-means clustering algorithm, a widely used unsupervised machine learning technique, was applied to partition the AE data into four clusters. The steps involved in the *k*-means algorithm are as follows: (1) Randomly select k initial cluster centroids; (2) Calculate the distance from each sample in the dataset to the centroid of k clusters, and assign it to the class corresponding to the centroid of the cluster with the smallest distance; (3) Recalculate the centroids of the clusters by taking the mean of all data points assigned to each cluster; (4) Repeat the process until convergence is achieved. The *k*-means clustering results, based on PF and PA, identified four clusters, each corresponding to a specific damage mode. The clustering results are shown in Fig. 5.

2.3.4. Damage modes identification

Based on the microscopic damage morphology of the fracture surface shown in Fig. 6(a)–(d), four distinct damage modes including matrix cracking, fiber pull-out, interface debonding, and fiber breakage could be identified. The specimen fracture surface was located at the overlap of yarns, as shown in Fig. 6(e)–(f). By correlating these damage modes with the materials' damage mechanisms, each frequency band should be associated with a specific damage mode.

It is believed that there is apparent correlation among the typical frequency, relaxation time of AE event and the material density. As mentioned in Ref. [10], this correlation can be defined as:

$$F \sim 1 / T \sim sqrt(E / \rho) \tag{4}$$

where F is the intrinsic frequency of AE event, T is the relaxation time of AE event, and E and ρ are the elastic modulus and density of the material respectively.

Fig. 7(a) shows the AE events over time, while Fig. 7(b) illustrates the accumulation of AE absolute energy for different modes with strain. Due to the presence of microcracks and voids, matrix cracking dominated the failure process in the initial loading stage. As shown in Fig. 7, it was evident that Cluster 1 exhibited the fastest rate of AE cumulative absolute energy in the initial loading stage, indicating that it corresponded to matrix cracking. As cracks propagated to the fiber-matrix interface, they induced interface debonding and slipping, rather than fiber breakage, if the interface was sufficiently weak. As shown in Fig. 7(a), with continued loading, the frequency range of Cluster 2 expanded, and a lower-frequency boundary was observed, indicating fiber slipping and interphase degradation during cyclic loading condition. Thus, Cluster 2 and Cluster 3 were associated with fiber pull-out and interface debonding, respectively. Finally, Cluster 4, which appeared later in the loading process, was characterized by low AE cumulative absolute energy and a broad frequency band. Some studies have shown that the frequency of fiber breakage exceeded 300 kHz [31]. Therefore, the high-frequency signals in Cluster 4 correspond to fiber breakage.

In summary, four damage modes can be correlated and mapped with the frequency ranges: matrix cracking was in the range of [0–180 kHz], fiber pull-out was in the range of [180–290 kHz], interface debonding was in the range of [290–390 kHz], and fiber breakage was in the range of [430–900 kHz]. Similar conclusions also can be found in other studies [29]. Furthermore, based on these results, an in-deep analysis of the composites' damage process will be presented in Section 4.2.

3. Meso-scale finite element model

In this section, a meso-scale finite element model was developed to capture the complex mechanical behavior of 2D C/SiC composites under cyclic loading, aiming to simulate the progressive damage accumulation and account for the interactions between fiber bundles and SiC matrix. The progressive damage models were introduced to describe the initiation and evolution of damage mechanisms such as matrix cracking, fiber pull-out, interface debonding, and fiber breakage. Following this, the meso-scale RVC model was presented to predict the nonlinear mechanical behavior of C/SiC composites under cyclic loading condition.

3.1. Progressive damage model

3.1.1. Constitutive model

The mechanical response of C/SiC composite under incremental loading/unloading process was predicted with the RVC model. For fiber

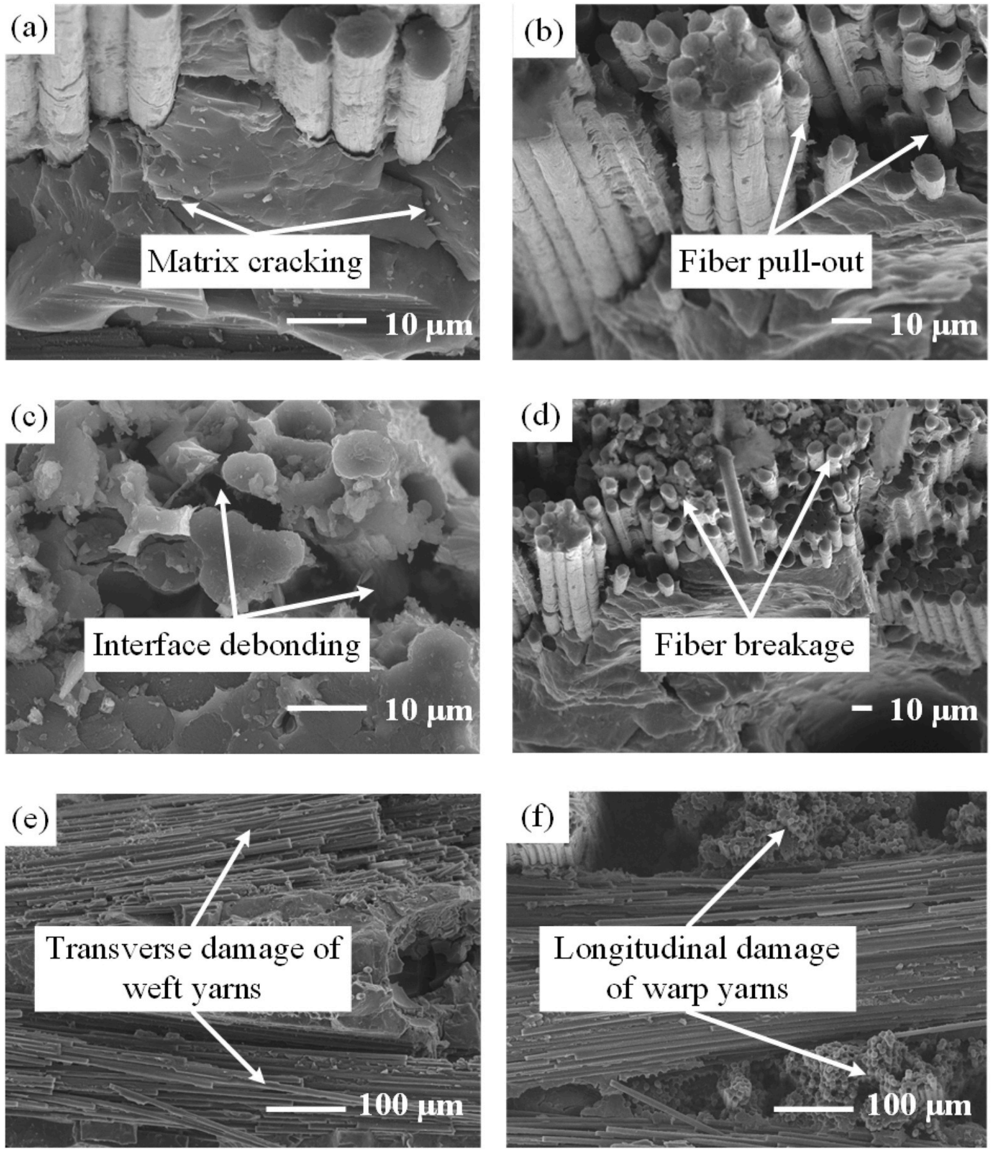

Fig. 6. Microscopic damage morphology of specimen fracture surface: (a) matrix cracking, (b) fiber pull-out, (c) interface debonding, (d) fiber breakage, (e) transverse damage of weft yarns and (f) longitudinal damage of warp yarns.

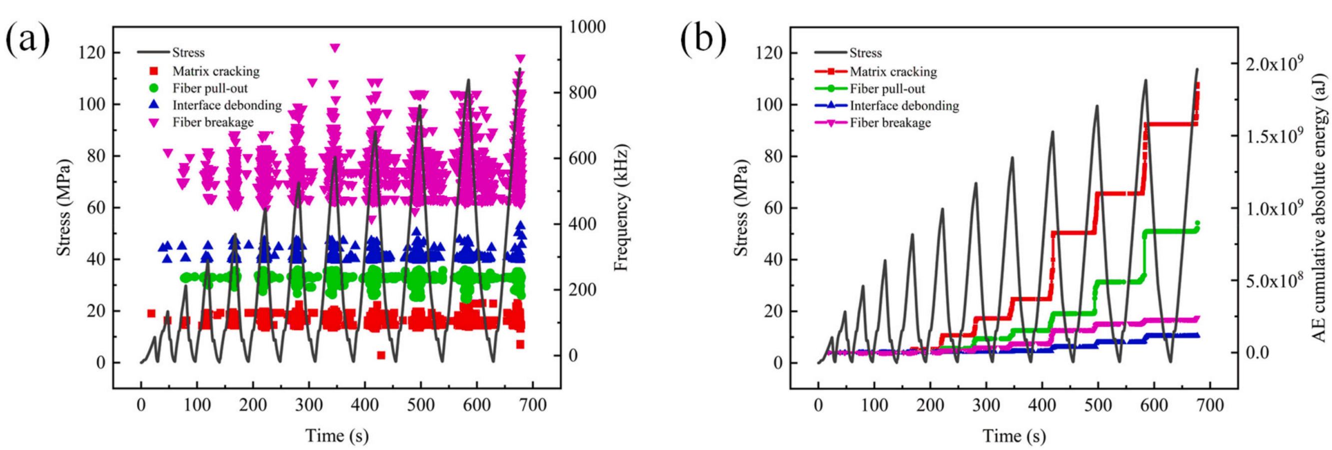

Fig. 7. Damage modes identification based on testing results of the typical specimen: (a) AE events over time and (b) AE cumulative absolute energy for different damage modes with strain.

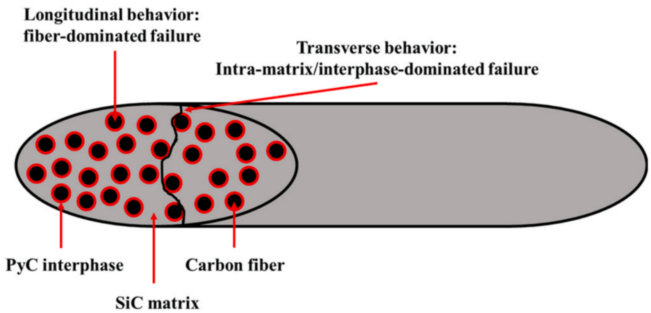

Fig. 8. Crack in the fiber bundle of CMCs [33].

bundles, typical failure modes in fiber bundles may be broadly classified into fiber-dominated failure modes (fiber breakage/pull-out) and intra-matrix/interphase-dominated failure modes (intra-matrix cracking/interface debonding) [29,32,33], as shown in Fig. 8. To describe the nonlinear behavior of CMCs' fiber bundles, a coupled anisotropic elastic-plastic damage constitutive model was proposed. Transverse damage was used to describe intra-matrix cracking and interface debonding. As to the longitudinal behavior of fiber bundles, we further considered the inelastic deformation and modulus degradation of fiber bundles, which were caused by fiber pull-out. Thus, in this study, inelastic deformation was used to describe fiber pull-out, and longitudinal damage was used to describe fiber breakage. For the matrix, considering SiC matrix was brittle, a linear elastic damage model was adopted.

For fiber bundle, the total strain tensor $\boldsymbol{\varepsilon}_f$ is decomposed into an elastic part $\boldsymbol{\varepsilon}_f^e$ and a plastic part $\boldsymbol{\varepsilon}_f^p$:

$$\boldsymbol{\varepsilon}_f = \boldsymbol{\varepsilon}_f^e + \boldsymbol{\varepsilon}_f^p \tag{5}$$

The constitutive equation of the fiber bundle can be defined as follows:

$$\boldsymbol{\sigma}_f = \boldsymbol{C}_f(\boldsymbol{d}) : \boldsymbol{\varepsilon}_f^e \tag{6}$$

where $\boldsymbol{\sigma}_f$ is the Cauchy nominal stress tensor, $\boldsymbol{C}_f(\boldsymbol{d})$ is the damaged stiffness matrix and $\boldsymbol{d}$ is the damage tensor. The effective stress tensor $\widetilde{\boldsymbol{\sigma}}$ can be defined as:

$$\widetilde{\boldsymbol{\sigma}}_f = \boldsymbol{C}_{f0} : \boldsymbol{\varepsilon}_f^e \tag{7}$$

where $\boldsymbol{C}_{f0}$ is the undamaged stiffness matrix. The effective stress tensor can be derived by nominal stress tensor, as follows:

$$\widetilde{\boldsymbol{\sigma}}_f = \boldsymbol{M}_f(\boldsymbol{d}) : \boldsymbol{\sigma}_f \tag{8}$$

where $\boldsymbol{M}_f(\boldsymbol{d})$ is the Matzenmiller damage tensor and can be written as follows:

$$\boldsymbol{M}_f(\boldsymbol{d}) = diag\left\{\frac{1}{1-d_{f,1}}, \frac{1}{1-d_{f,2}}, \frac{1}{1-d_{f,3}}, \frac{1}{1-d_{f,4}}, \frac{1}{1-d_{f,5}}, \frac{1}{1-d_{f,6}}\right\} \tag{9}$$

where $d_{f,i}$ is the damage variable. It is postulated that the damage variables $d_{f,4}$, $d_{f,5}$ and $d_{f,6}$ are not independent, and can be expressed as a function of the remaining variables [34]:

$$\begin{aligned} d_{f,4} &= 1-(1-d_{f,1})(1-d_{f,2}) \\ d_{f,5} &= 1-(1-d_{f,1})(1-d_{f,3}) \\ d_{f,4} &= 1-(1-d_{f,2})(1-d_{f,3}) \end{aligned} \tag{10}$$

In this model, the effective stress tensor is used to couple the plasticity and damage effects into a single constitutive equation. This approach allows for the independent modeling of damage and plastic deformation. The anisotropic damage variable accounts for softening and the decrease in elastic stiffness in different directions, while plasticity governs irreversible strain development. Plastic deformation, driven by effective stresses, is modeled separately from damage.

The Hill criterion [35] is applied as the yield function:

$$\begin{aligned}\Phi(\widetilde{\boldsymbol{\sigma}}_f, \overline{\sigma}) = {}& C_1(\widetilde{\sigma}_{f,11} - \widetilde{\sigma}_{f,22})^2 + C_2(\widetilde{\sigma}_{f,22} - \widetilde{\sigma}_{f,33})^2 + C_3(\widetilde{\sigma}_{f,33} - \widetilde{\sigma}_{f,11})^2 + \\ & C_4\widetilde{\sigma}_{f,12}^2 + C_5\widetilde{\sigma}_{f,13}^2 + C_6\widetilde{\sigma}_{f,23}^2 - \overline{\sigma}^2 = f(\widetilde{\boldsymbol{\sigma}}_f) - \overline{\sigma}^2\end{aligned} \tag{11}$$

where $\overline{\sigma}$ is the non-dimensional relative yield stress and the material parameters C_i can be determined by:

$$\begin{aligned} C_1 &= \frac{1}{2}\left(\frac{1}{(\overline{\sigma}_{f,11}^0)^2} + \frac{1}{(\overline{\sigma}_{f,22}^0)^2} - \frac{1}{(\overline{\sigma}_{f,33}^0)^2}\right); \\ C_2 &= \frac{1}{2}\left(-\frac{1}{(\overline{\sigma}_{f,11}^0)^2} + \frac{1}{(\overline{\sigma}_{f,22}^0)^2} + \frac{1}{(\overline{\sigma}_{f,33}^0)^2}\right); \\ C_3 &= \frac{1}{2}\left(\frac{1}{(\overline{\sigma}_{f,11}^0)^2} - \frac{1}{(\overline{\sigma}_{f,22}^0)^2} + \frac{1}{(\overline{\sigma}_{f,33}^0)^2}\right); \\ C_4 &= \frac{1}{(\overline{\sigma}_{f,12}^0)^2}; C_5 = \frac{1}{(\overline{\sigma}_{f,13}^0)^2}; C_6 = \frac{1}{(\overline{\sigma}_{f,23}^0)^2}; \end{aligned} \tag{12}$$

where $\overline{\sigma}_{f,ij}^0$ represents the initial yield strength of the fiber bundle.

The yield criterion can also be reformulated in matrix/vector notation:

$$\Phi(\widetilde{\boldsymbol{\sigma}}_f, \overline{\sigma}) = \frac{1}{2}\widetilde{\boldsymbol{\sigma}}_f^T \boldsymbol{P} \widetilde{\boldsymbol{\sigma}}_f - \overline{\sigma}^2 \tag{13}$$

where

$$\boldsymbol{P} = 2\begin{bmatrix} C_1+C_3 & -C_1 & -C_3 & 0 & 0 & 0 \\ -C_1 & C_1+C_2 & -C_2 & 0 & 0 & 0 \\ -C_3 & -C_2 & C_2+C_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & C_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & C_5 & 0 \\ 0 & 0 & 0 & 0 & 0 & C_6 \end{bmatrix} \tag{14}$$

Associated flow rule is used and the plastic potential function $g(\widetilde{\boldsymbol{\sigma}}_f)$ and the flow direction $\boldsymbol{N}$ can be defined as:

$$g(\widetilde{\boldsymbol{\sigma}}_f) = \frac{1}{2}\widetilde{\boldsymbol{\sigma}}_f^T \boldsymbol{P} \widetilde{\boldsymbol{\sigma}}_f \tag{15}$$

$$\boldsymbol{N} = \frac{\partial g(\widetilde{\boldsymbol{\sigma}}_f)}{\partial \widetilde{\boldsymbol{\sigma}}_f} = \boldsymbol{P}\widetilde{\boldsymbol{\sigma}}_f \tag{16}$$

The plastic strain rate is given by:

$$\dot{\boldsymbol{\varepsilon}}_f^p = \dot{\gamma}\boldsymbol{N} \tag{17}$$

where $\dot{\gamma}$ is the rate of the non-negative plastic multiplier.

Based on the plastic work equivalence principle, the rate of the plastic work is defined as:

$$\dot{W}^p = \widetilde{\boldsymbol{\sigma}}_f \dot{\boldsymbol{\varepsilon}}_f^p = \overline{\sigma}_f^0 \overline{\sigma} \dot{\overline{\varepsilon}}_f^p \tag{18}$$

where $\dot{\overline{\varepsilon}}_f^p$ is the rate of the equivalent plastic strain, which can be found by substituting Eq. (11), Eq. (16) and Eq. (17) into Eq. (18) as follows:

$$\dot{\overline{\varepsilon}}_f^p = \dot{\gamma}\widetilde{\boldsymbol{\sigma}}_f \frac{\partial g(\widetilde{\boldsymbol{\sigma}}_f)}{\partial \widetilde{\boldsymbol{\sigma}}_f} \frac{1}{\sqrt{f(\widetilde{\boldsymbol{\sigma}}_f)}} \frac{1}{\overline{\sigma}_f^0} \tag{19}$$

SiC matrix is isotropic material and the constitutive equation of the

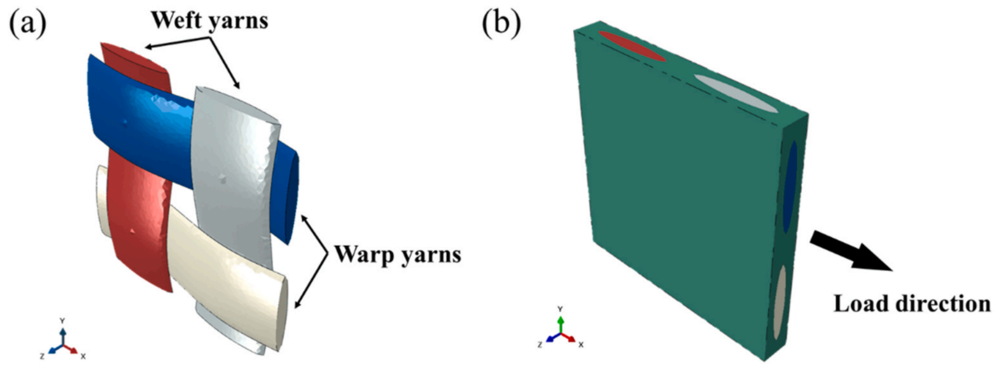

Fig. 9. The meso-scale finite element model: (a) weft yarns and warp yarns, and (b) load direction.

SiC matrix is defined as follows:

$$\boldsymbol{\sigma}_m = \boldsymbol{C}_m(d) : \boldsymbol{\varepsilon}_m \tag{20}$$

where $\boldsymbol{\sigma}_m$ is the Cauchy nominal stress tensor of the SiC matrix and $\boldsymbol{C}_m(d)$ is the damaged stiffness matrix. The effective stress tensor $\widetilde{\boldsymbol{\sigma}}_m$ of the SiC matrix is defined as

$$\widetilde{\boldsymbol{\sigma}}_m = \boldsymbol{C}_{m0} : \boldsymbol{\varepsilon}_m \tag{21}$$

where $\boldsymbol{C}_{m0}$ is the undamaged stiffness matrix.

3.1.2. Damage initiation criteria

For the fiber bundle, the 3D Hashin failure criterion [36] is adopted, which is derived from the effective stress tensor as follows:

Table 1
Material properties of C/SiC fiber bundle [13,43,44].

Fiber bundle	$E_{f,11}$ (GPa)	$E_{f,22(33)}$ (GPa)	$G_{f,12(13)}$ (GPa)	$G_{f,23}$ (GPa)	$\nu_{f,12(13)}$
	126	60	24	44.4	0.26
	$X_{f,11t}$ (MPa)	$X_{f,11c}$ (MPa)	$X_{f,22(33)t}$ (MPa)	$X_{f,22(33)c}$ (MPa)	$X_{f,12(13)}$ (MPa)
	238.91	409.40	40	120.19	45
	$X_{f,23}$ (MPa)	$\overline{\sigma}^0_{f,11}$ (MPa)	$\overline{\sigma}^0_{f,22(33)}$ (MPa)	$\overline{\sigma}^0_{f,12(13)}$ (MPa)	$\overline{\sigma}^0_{f,23}$ (MPa)
	40	40	1000000	1000000	1000000

$$\begin{aligned}
&\phi_{f,11t} = \sqrt{\left(\frac{\widetilde{\sigma}_{f,11}}{X_{f,11t}}\right)^2 + \alpha\left(\frac{\widetilde{\sigma}_{f,12}}{X_{f,12}}\right)^2 + \beta\left(\frac{\widetilde{\sigma}_{f,31}}{X_{f,31}}\right)^2} \geq 1, \widetilde{\sigma}_{f,11} \geq 0\\
&\phi_{f,11c} = \sqrt{\left(\frac{\widetilde{\sigma}_{f,11}}{X_{f,11c}}\right)^2} \geq 1, \widetilde{\sigma}_{f,11} < 0\\
&\phi_{f,22(33)t} = \sqrt{\left(\frac{\widetilde{\sigma}_{f,22} + \widetilde{\sigma}_{f,33}}{X_{f,22(33)t}}\right)^2 + \frac{\left(\widetilde{\sigma}^2_{f,23} - \widetilde{\sigma}_{f,22}\widetilde{\sigma}_{f,33}\right)}{X^2_{f,23}} + \left(\frac{\widetilde{\sigma}_{f,12}}{X_{f,12}}\right)^2 + \left(\frac{\widetilde{\sigma}_{f,31}}{X_{f,31}}\right)^2} \geq 1, \widetilde{\sigma}_{f,22} + \widetilde{\sigma}_{f,33} \geq 0\\
&\phi_{f,22(33)c} = \sqrt{\frac{1}{X_{f,22(33)c}}\left[\left(\frac{X_{f,22(33)c}}{2X_{f,23}}\right)^2 - 1\right]\left(\widetilde{\sigma}_{f,22} + \widetilde{\sigma}_{f,33}\right) + \left(\frac{\widetilde{\sigma}_{f,22} + \widetilde{\sigma}_{f,33}}{2X_{f,23}}\right)^2 + \frac{\left(\widetilde{\sigma}^2_{f,23} - \widetilde{\sigma}_{f,22}\widetilde{\sigma}_{f,33}\right)}{X^2_{f,23}} + \left(\frac{\widetilde{\sigma}_{f,12}}{X_{f,12}}\right)^2 + \left(\frac{\widetilde{\sigma}_{f,31}}{X_{f,31}}\right)^2} \geq 1, \widetilde{\sigma}_{f,22} + \widetilde{\sigma}_{f,33} < 0
\end{aligned} \tag{22}$$

where the subscript t and c represent the stress states of tension and compression, and X is the strength in different directions.

For the matrix, the initiation damage criterion is based on the von Mises theory [37] as follows:

$$\begin{aligned}
&\phi_{m,t} = \sqrt{\frac{\widetilde{\sigma}^2_{m,11} + \widetilde{\sigma}^2_{m,22} + \widetilde{\sigma}^2_{m,33} - \widetilde{\sigma}_{m,11}\widetilde{\sigma}_{m,22} - \widetilde{\sigma}_{m,22}\widetilde{\sigma}_{m,33} - \widetilde{\sigma}_{m,11}\widetilde{\sigma}_{m,33} + 3\widetilde{\sigma}^2_{m,12} + 3\widetilde{\sigma}^2_{m,23} + 3\widetilde{\sigma}^2_{m,31}}{X^2_{m,t}}} \geq 1, \widetilde{I}_m = \widetilde{\sigma}_{m,11} + \widetilde{\sigma}_{m,22} + \widetilde{\sigma}_{m,33} \geq 0\\
&\phi_{m,c} = \sqrt{\frac{\widetilde{\sigma}^2_{m,11} + \widetilde{\sigma}^2_{m,22} + \widetilde{\sigma}^2_{m,33} - \widetilde{\sigma}_{m,11}\widetilde{\sigma}_{m,22} - \widetilde{\sigma}_{m,22}\widetilde{\sigma}_{m,33} - \widetilde{\sigma}_{m,11}\widetilde{\sigma}_{m,33} + 3\widetilde{\sigma}^2_{m,12} + 3\widetilde{\sigma}^2_{m,23} + 3\widetilde{\sigma}^2_{m,31}}{X^2_{m,c}}} \geq 1, \widetilde{I}_m = \widetilde{\sigma}_{m,11} + \widetilde{\sigma}_{m,22} + \widetilde{\sigma}_{m,33} \leq 0
\end{aligned} \tag{23}$$

Table 2
Material properties of SiC matrix [29,44].

Matrix	E_m (GPa)	G_m (GPa)	ν_m	$X_{m,t(c)}$(MPa)
	105	38	0.39	100

Table 3
The comparison of the main mechanical data of C/SiC composites obtained by experiments and simulations.

	Experiment	Simulation	Error
Initial elastic modulus, $E_{initial}$ (GPa)	86.25 ± 4.54	81.93	−5.00 %
Ultimate strength, $\sigma_{ultimate}$ (MPa)	113.52 ± 2.80	117.61	3.60 %
Matrix cracking stress, σ_{mc} (MPa)	31.69 ± 10.40	48.00	51.47 %

where $X_{m,t}$ and $X_{m,c}$ represent the tensile strength and compressive strength of the matrix, respectively.

3.1.3. Damage evolution law

The exponential damage law is adopted to characterize the softening process once the damage initiation criterion is met. The exponential damage law [38] is defined as follows:

$$\begin{aligned} d_{f,I} &= 1 - \frac{1}{r_{f,I}}\exp\left[A_{f,I}\left(1 - r_{f,I}\right)\right] (I = 11t, 11c, 22t, 22c, 33t, 33c) \\ d_m &= 1 - \frac{1}{r_m}\exp[A_m(1 - r_m)] \end{aligned} \tag{24}$$

where $A_{f,I}$ represents the softening coefficient of the fiber bundle under different conditions and A_m represents the softening coefficient of the matrix. Their definitions can be found in Ref. [39].

In this study, by deriving the anisotropic elastic-plastic damage constitutive model and consistent tangent stiffness matrix, an implicit solution method was employed to achieve accurate and efficient solutions for predicting the nonlinear mechanical response of C/SiC composites. The models were implemented in ABAQUS software with a user subroutine UMAT. More details of the damage models about the expressions and numerical implementation could be found in Appendix B.

3.2. Meso-scale finite element model

The macroscopic properties of the composites could be obtained by analyzing the mechanical behavior of RVC finite element model. The typical structure of C/SiC composites was obtained by X-ray computed tomography scanning in our previous work [40]. The RVC model size was 3.5 mm × 3.5 mm × 0.4334 mm. The fiber cross-section was ellipse-shaped and the length of its major axis and minor axis was 1.28 mm and 0.20 mm respectively. The fiber volume fraction was 40 %. The meso-scale finite element model was established with TexGen software and 116,724 C3D4 elements were used to discretize the whole model. The periodic boundary conditions were applied to ensure the compatibility of deformations and continuity of stress [41]. The meso-scale finite element model is shown in Fig. 9.

In this study, the following assumptions were made: inelastic deformation of the fiber bundle occurred only when it was subjected to axial tensile loading, which resulted from internal matrix cracking, fiber slippage, and other factors. Inelastic deformation was not considered in the fiber bundle along other directions. Consequently, based on the experiment results, the axial yield strength of the fiber bundle was set to 40 MPa, and the hardening curve for the fiber bundle could be referred to Ref. [29]. The yield strength of the fiber bundle in all other directions was assumed to be infinite. Additionally, there existed tension-compression distinction along the axial direction of the fiber bundle [13,42]. However, this effect was also neglected in the current model. Future work will focus on improving the model to incorporate the tensile-compressive anisotropy of the fiber bundle. The material properties of the C/SiC fiber bundle and the SiC matrix are shown in Tables 1 and 2 respectively.

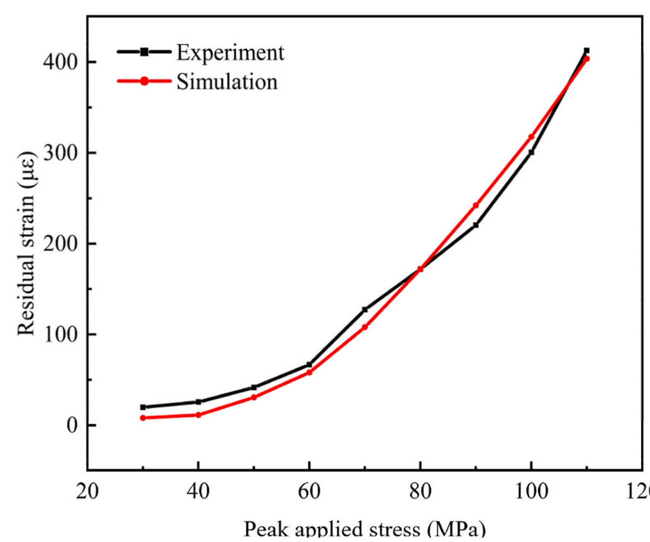

Fig. 11. Evolution of the residual strains of 2D-C/SiC composites under cyclic tensile load.

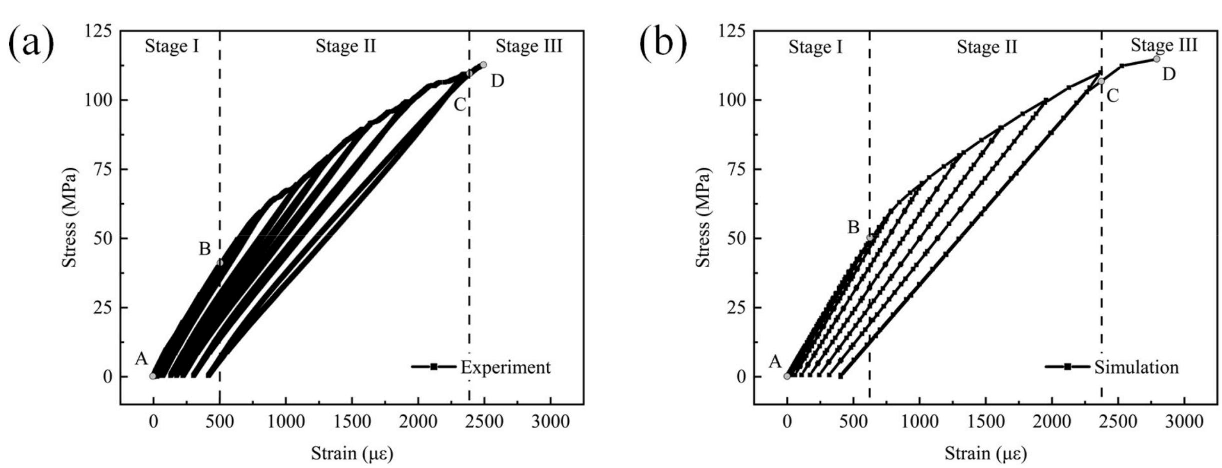

Fig. 10. The typical loading-unloading tensile stress-strain curve of C/SiC composites: (a) experiment and (b) simulation.

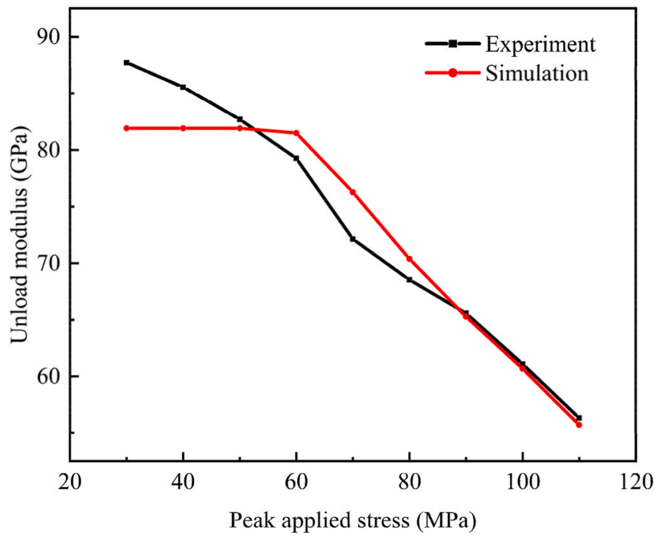

Fig. 12. Evolution of the unloading modulus of 2D-C/SiC composites under cyclic tensile load.

4. Results and discussion

In this section, the results of experimental and numerical investigations on the mechanical behavior and damage evolution of 2D C/SiC composites under cyclic loading are presented. The key findings are discussed in detail, focusing on the observed damage mechanisms, the evolution of material properties, and performance trends under cyclic loading. The experimental results were compared with the simulation results across critical aspects: stress-strain curves, residual strain, unloading modulus, damage evolution process and failure modes, to validate the proposed meso-scale finite element model.

4.1. Stress-strain curves

Table 3 presents the mechanical parameters of the composites obtained from both experimental and computational results, including the initial stiffness, strength, and matrix cracking stress. According to the experimental results, the initial stiffness of the composites was 86.25 GPa, the strength was 113.52 MPa, and the matrix cracking stress was 31.69 MPa. In comparison, the computational results yield an initial stiffness of 81.93 GPa, strength of 117.61 MPa, and matrix cracking stress of 48 MPa. Their discrepancies between the experimental and computational results were −5.0 %, 3.60 % and 51.47 %, respectively. It was noted that there was an error of up to 51.47 % about matrix cracking stress. Three factors contributed to the apparent error: (i) Clamping condition. The linear segments of the composites' stress-strain curves were relatively short, and the test results were easily impacted by the specimen clamping state. (ii) Measurement error. A tangent line was drawn at the linear segment, and the proportional limit was identified and regarded as the matrix cracking stress. However, this method could be prone to errors influenced by human factors. To minimize such impacts, we determined the matrix cracking stress by averaging multiple measurements. (iii) Effect of microcracks. The simulation model ignored the material's local microstructural features. By analyzing the experiment results, we found that at the initial loading stage, the material's tangent stiffness continued to decrease, but no significant acoustic emission signals or residual deformation were detected. This suggested that the opening/closure of inherent microcracks introduced the nonlinear mechanical response in this stage. However, the mesoscopic

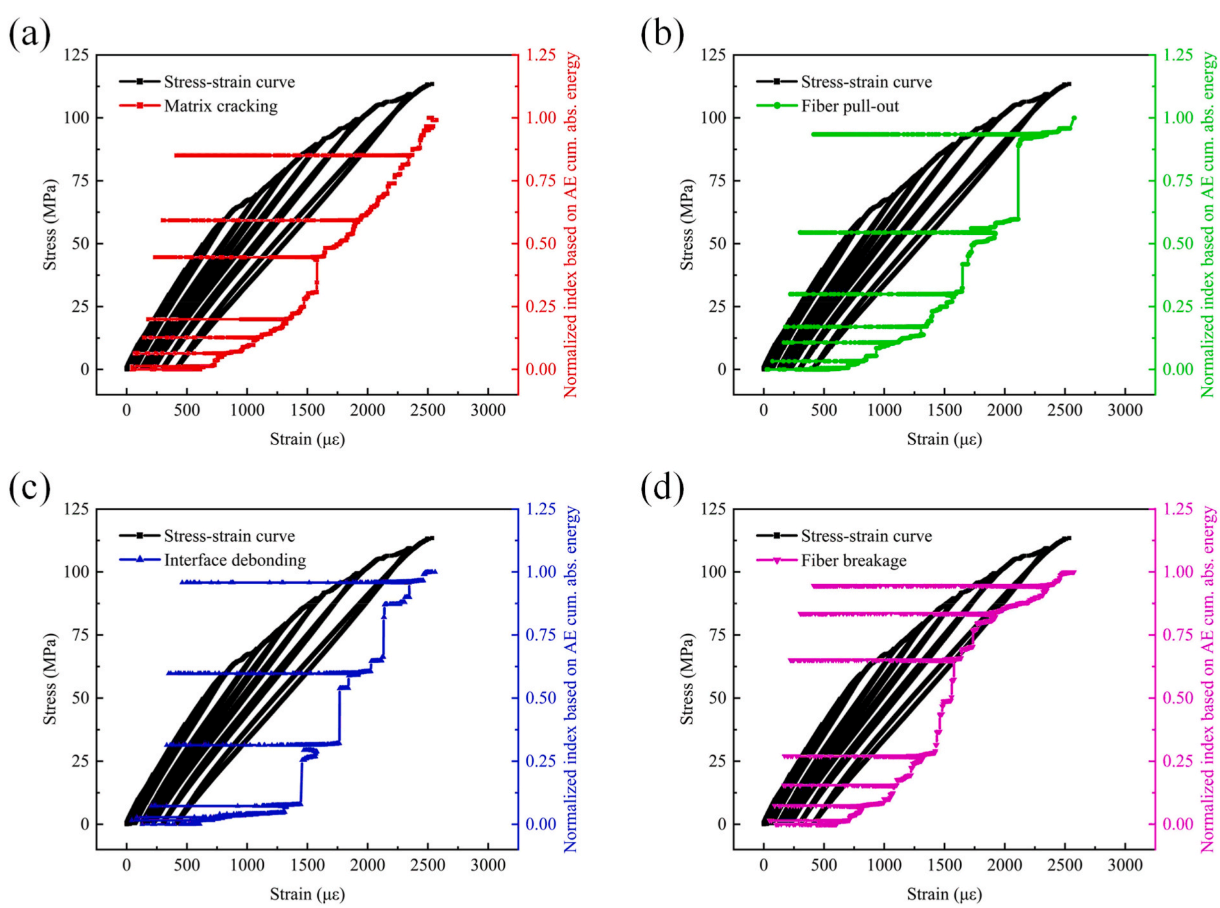

Fig. 13. Damage process analysis based on AE results: (a) matrix cracking, (b) fiber pull-out, (c) interface debonding and (d) fiber breakage.

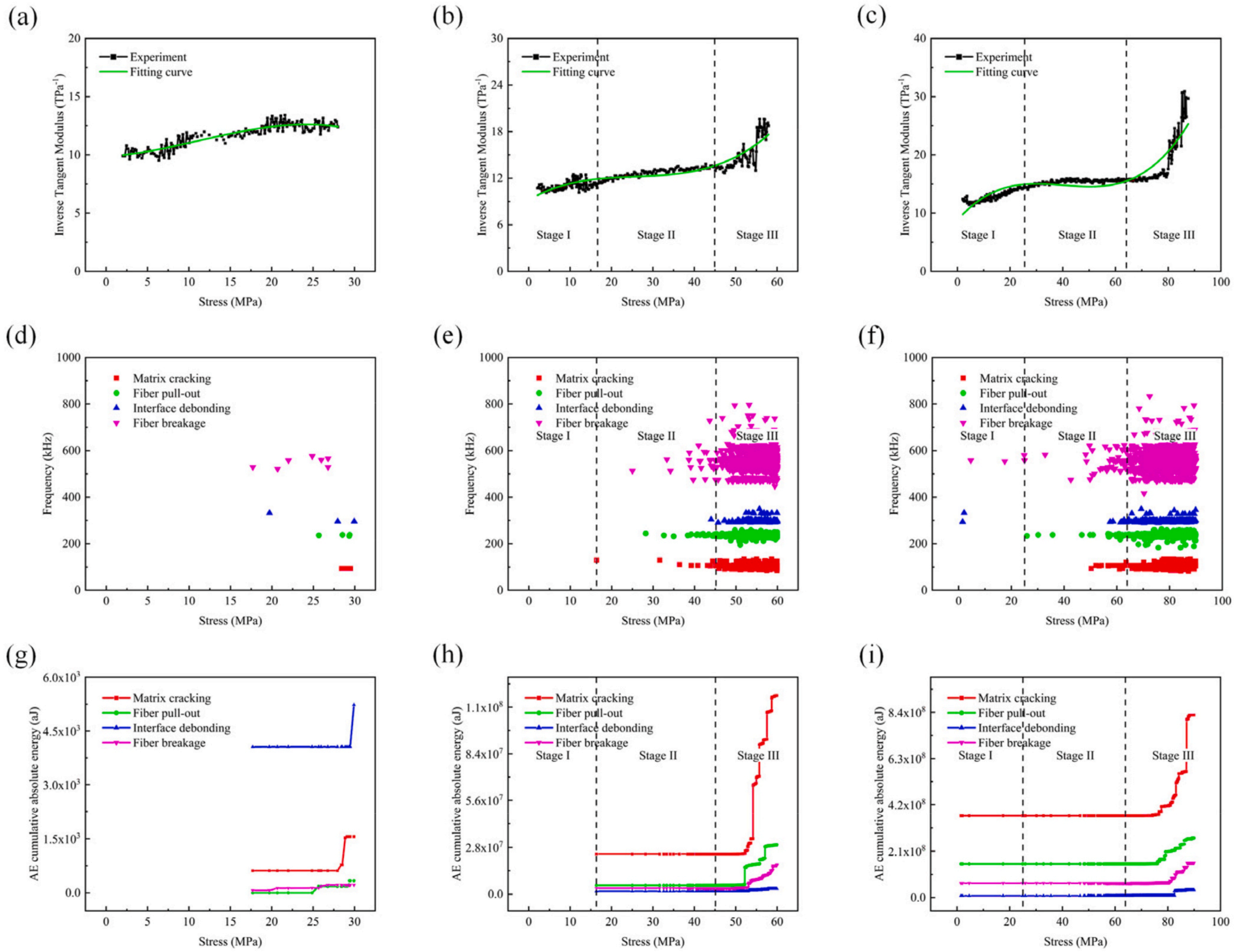

Fig. 14. Inverse tangent modulus response and AE results in reloading processes: (a)–(c) inverse tangent modulus response in 3rd, 6th and 9th reloading process, (d)–(f) acoustic emissions in 3rd, 6th and 9th reloading process and (g)–(i) AE cumulative absolute energy of four damage modes in 3rd, 6th and 9th reloading process.

finite element model did not account for the material's local microstructural features, which could not obtain accurate mechanical responses in the initial loading stage. In summary, the proposed model could describe the nonlinear mechanical behavior well and the simulation errors were within an acceptable range.

Fig. 10 presents the loading-unloading tensile stress-strain curve of C/SiC composites obtained by experiments and simulations. The experimental stress-strain curve closely matched the computational stress-strain curve, demonstrating the validity of the computational method. As shown in Fig. 10, the stress-strain response could be categorized into three distinct stages. At the first stage, the C/SiC composites exhibited elastic behavior with the strain smaller than 500 με. At the second stage, due to matrix cracking and inter-tow matrix cracking, a nonlinear response and a significant decrease in modulus were observed with the strain between 500 με and 2500 με. At the third stage, universal fiber breakage led to composite failure rapidly with the strain above 2500 με.

Figs. 11 and 12 present the evolution of residual strains and unloading modulus for 2D woven composites, respectively. The simulation results showed a good agreement with the experimental data. As shown in Fig. 11, residual strain increased continuously with the peak applied stress. When the peak stress was below 40 MPa, residual strain increased gradually. However, there was a rapid increase in residual strain when the peak stress exceeded 40 MPa. A similar trend was observed in the simulation results. The residual strain in CMCs could be attributed to several factors: (i) the release of residual stress, (ii) partial irreversible sliding arising from the various energy dissipative frictional mechanisms and (iii) a mechanical impediment of complete crack closure. Fig. 12 shows the evolution of the unloading modulus under cyclic loading. Based on the experimental data, it could be concluded that when the peak applied stress exceeded 40 MPa, the unloading modulus decreased rapidly. A similar trend was observed in the simulation results. Nevertheless, there were some discrepancies between the experimental and numerical results for the unloading modulus when the applied stress was below 40 MPa. At the initial loading stage, the nonlinear behavior of the composites was primarily governed by the opening and closure of internal microcracks. However, the proposed model was unable to fully capture this effect of these microcracks. These factors resulted in the apparent discrepancies, particularly at initial loading stage.

4.2. Damage process analysis based on AE results

In this section, the damage evolution in the 2D C/SiC composites under cyclic loading was investigated through AE monitoring. AE signals provided valuable insights into the internal damage mechanisms. The damage evolution was quantified using a normalized damage index, which was defined as the ratio of the AE cumulative absolute energy at the *i*th second to the maximum cumulative absolute energy observed before failure. This approach allowed for a real-time assessment of the

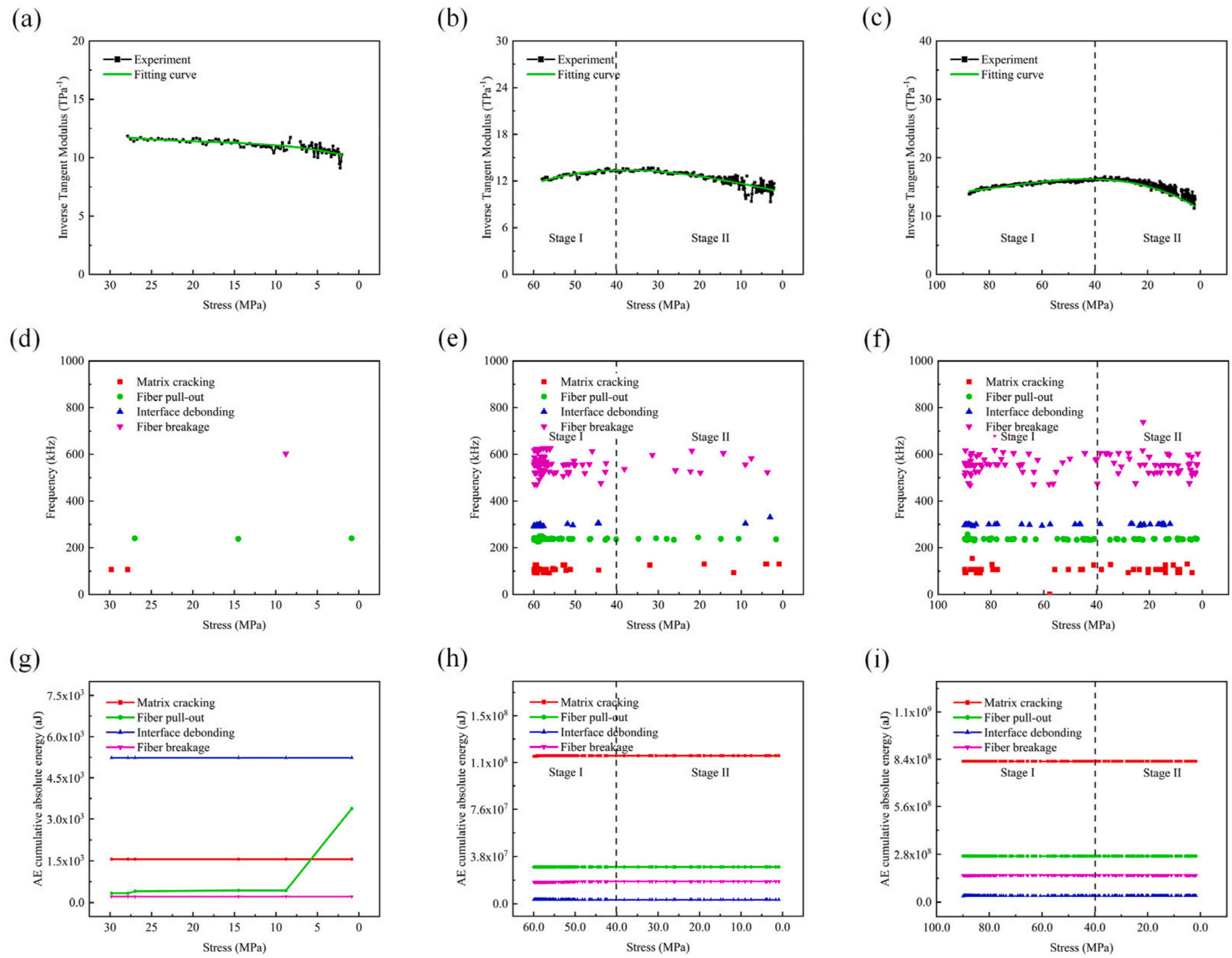

Fig. 15. Inverse tangent modulus response and AE results in unloading processes: (a)–(c) inverse tangent modulus response in 3rd, 6th and 9th unloading process, (d)–(f) acoustic emissions in 3rd, 6th and 9th unloading process and (g)–(i) AE cumulative absolute energy of four damage modes in 3rd, 6th and 9th unloading process.

damage state throughout the loading-unloading cycles.

Fig. 13 presents the damage process of the composites based on AE results. As shown in Fig. 13(a), at the initial loading stage, the matrix did not experience any damage. Once the load exceeded the matrix cracking stress, damage occurred and accumulated in the SiC matrix. When the strain was less than 750 με, matrix damage was minimal. Between the strain range of 750 με to 1350 με, matrix damage accumulated slowly. At the strain of 1350–1650 με (with an external load of approximately 80 MPa), the matrix damage suddenly and rapidly increased, which indicated that local microcracks propagated fast and even merged to form main cracks. When the strain exceeds 1650 με, matrix damage evolved rapidly, and the damage rate increased significantly as the material approaches its ultimate strength. As shown in Fig. 13(b), when the strain was less than 750 με, the AE signals of fiber pull-out were seldom detected. Once the strain exceeds 750 με, the AE signals of fiber pull-out became more frequent. As presented in Fig. 13(c), when the strain was less than 1500 με, the AE signals of interface debonding were sparse and increase slowly. Once the strain exceeded 1500 με, these signals increased dramatically, indicating that interface debonding occurred predominantly in the high-stress phase. As shown in Fig. 13(d), when the strain was less than 750 με, the AE signals of fiber breakage were almost nonexistent. Between the strain range of 750 με to 1300 με, these signals accumulated slowly. However, at a strain of 1300–1550 με, these signals increased rapidly. This increase of signals indicated that local failure of the composites happened, which was closely related to the in-situ strength of the fibers. When the strain exceeded 1550 με, the AE signals of fiber breakage continued to accumulate. Localized fiber breakage in C/SiC composites could trigger global structural failure, particularly under cyclic loading conditions.

The stress-strain curve showed the mechanical behaviors of the composites in macroscopic, which were conducted by internal damage matters in microscopic. In this section, the damage process of C/SiC composites under certain loading/unloading process was studied by analyzing the stress-strain curves and AE results. Inverse tangent modulus showed the damage extent of the composites instantly [45]. The inverse tangent modulus and the AE results in reloading processes and unloading processes are shown in Figs. 14 and 15, respectively.

The inverse tangent modulus and the AE results in reloading processes are analyzed, as shown in Fig. 14. When the peak applied stress was below 40 MPa, as shown in Fig. 14(a), inverse tangent modulus was between 9 TPa^{-1} and 14 TPa^{-1}, and rare acoustic emission was detected as shown in Fig. 14(d) and (g). The change of Inverse tangent modulus might be contributed to the reverse slip with matrix crack opening. When the peak applied stress was above 40 MPa, the mechanical behavior in reloading process could be divided into three stages, as shown in Fig. 14(b) and (c). At the first stage, from the beginning of the reloading process, inverse tangent modulus increases slowly, which was caused by the reverse slip with matrix crack opening. No acoustic

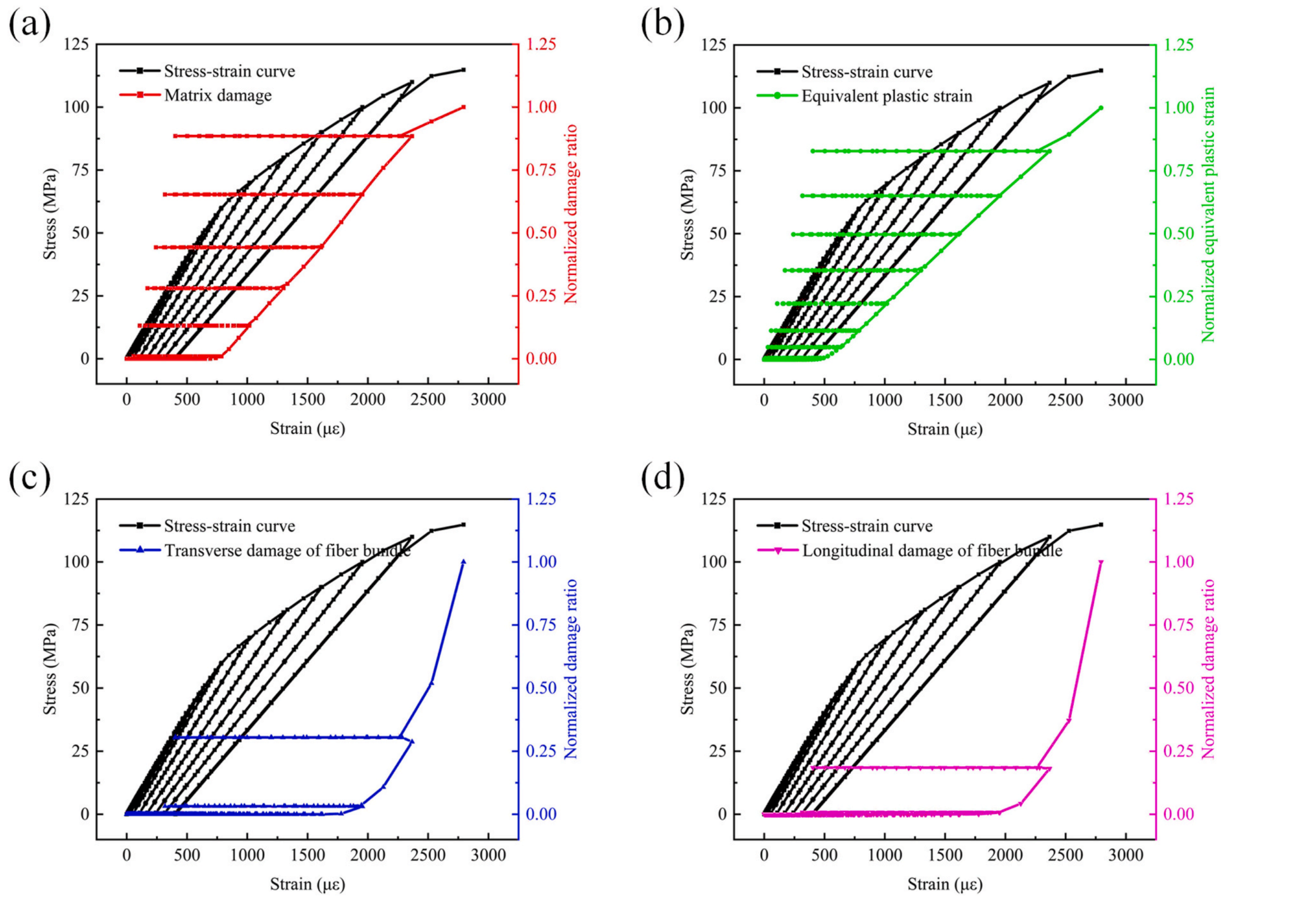

Fig. 16. Damage process analysis based on simulation: (a) matrix damage, (b) equivalent plastic strain, (c) transverse damage and (d) longitudinal damage.

emission was detected in this stage. At the second stage, inverse tangent modulus nearly remained constant. The acoustic emissions of matrix cracking, fiber pull-out and fiber breakage were seldom detected. At the third stage, near the peak applied stress, inverse tangent modulus increased rapidly. A great sum of acoustic emissions were detected. Inverse tangent modulus-stress curves in reloading processes could be fitted by cubic polynomial.

The inverse tangent modulus and the AE results in unloading processes are analyzed, as shown in Fig. 15. When the peak applied stress was below 40 MPa, inverse tangent modulus in the unloading process was between 9 TPa^{-1} and 12 TPa^{-1}, and seldom acoustic emission was detected, as shown in Fig. 15(d) and (g). The change of inverse tangent modulus might be attributed to the reverse slip with matrix crack closure. When the peak applied stress was between 40 MPa and 80 MPa, the mechanical response in unloading process could be divided into two stages, as shown in Fig. 15(b). At the first stage, from the beginning of unloading process, the applied stress was near the peak applied stress and inverse tangent modulus increased slightly. Some acoustic emissions of all damage modes were detected but the cumulative absolute energy of every damage mode almost did not change. At the second stage, inverse tangent modulus decreased slowly and rare acoustic emission was detected. The change of inverse tangent modulus was contributed to reverse slip with matrix crack closure. When the peak applied stress was above 80 MPa, the mechanical behavior in unloading process could also be divided into two stages, as shown in Fig. 15(c). At the first stage, from the beginning of unloading process, inverse tangent modulus increased slightly which was caused by further damage of the composites. Some acoustic emissions of all damage modes were detected. At the second stage, inverse tangent modulus decreased slowly, which was mainly caused by matrix crack closure. Inverse tangent modulus-stress curves in unloading processes could be fitted by quadratic polynomial.

4.3. *Damage process analysis based on simulation*

In this section, the damage evolution in the 2D C/SiC composites under cyclic loading was investigated through simulation results to provide a comprehensive understanding of the complex mechanical behavior. The damage evolution was characterized by matrix damage, equivalent plastic strain, transverse damage and longitudinal damage. Additionally, the volume average of the damage variable or the equivalent plastic strain at *i*th increment was calculated first. Then the normalized index was defined by dividing the volume average at *i*th increment by the maximum of the volume average obtained when the composites failed.

Fig. 16 illustrates the evolution process of four damage modes based on simulation results. Fig. 16(a) shows the evolution of matrix damage. When the strain was below 750 με, the matrix damage remained relatively small. However, once the strain exceeded 750 με, matrix damage increased rapidly and accumulated progressively. As the external load further increased, the rate of matrix damage accumulation accelerated. The overall evolution of matrix damage predicted by the proposed model was in good agreement with the AE analysis results. Fig. 16(b) shows the evolution of the equivalent plastic strain. Inelastic deformation occurred in fiber bundles due to fiber pull-out. When the strain was below 500 με, the inelastic deformation of the fiber bundle was minimal. But after the strain exceeded this threshold, the inelastic deformation accumulated gradually, and the accumulation rate accelerated as the load increased. This trend matched the experimental results. Fig. 16(c) depicts the evolution of transverse damage in the fiber bundles. The first occurrence of transverse damage was observed around a strain of about 1750 με. With load increasing, the transverse damage accumulated

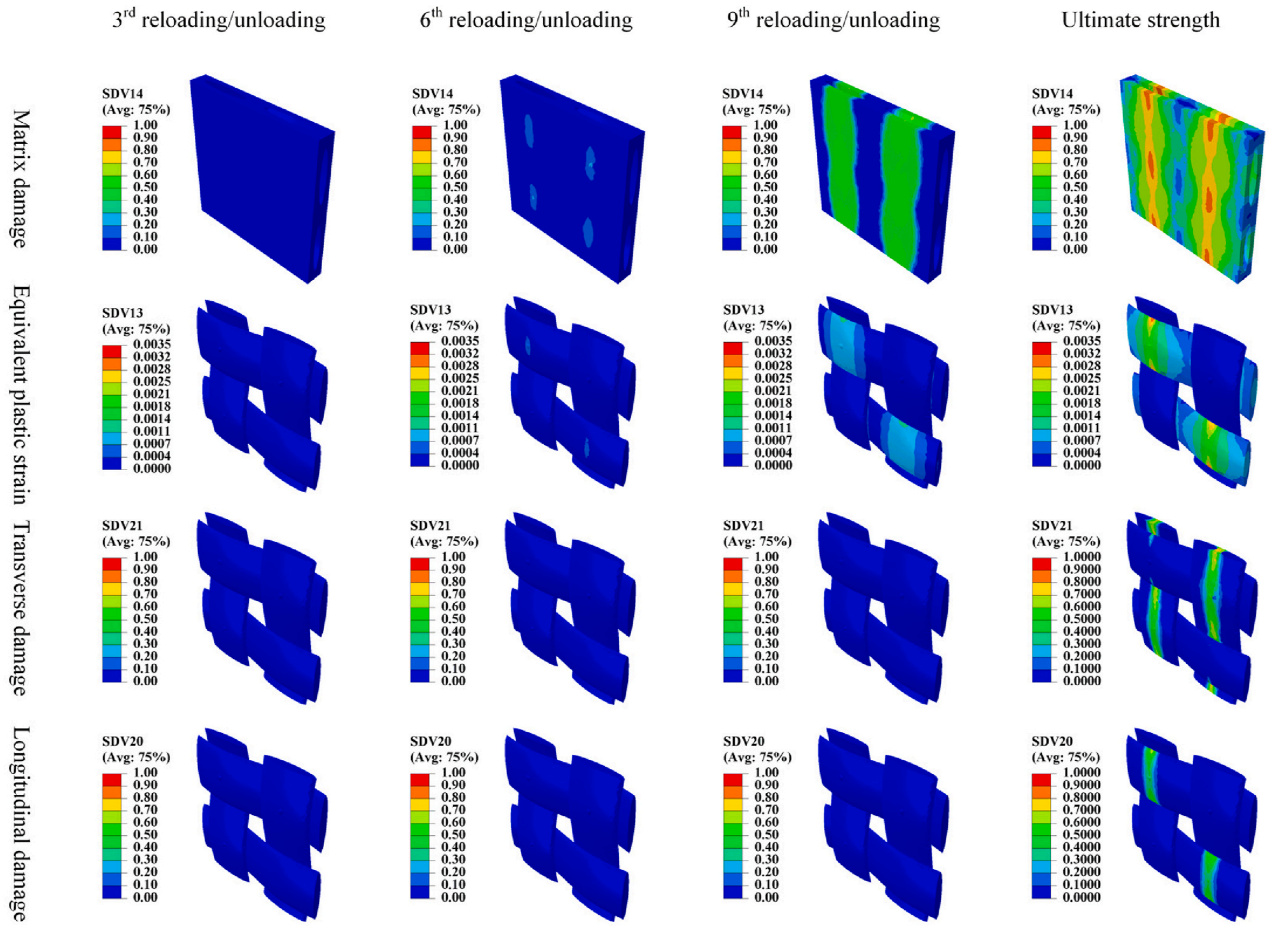

Fig. 17. Damage process analysis based on simulation: (a) matrix damage, (b) equivalent plastic strain, (c) transverse damage and (d) longitudinal damage.

slowly. When the strain exceeded 2300 με, the rate of transverse damage accumulation increased rapidly. Fig. 16(d) illustrates the evolution of longitudinal damage in the fiber bundles. Longitudinal damage happened at a strain of 1750 με. Between strains of 1750 με and 2000 με, the longitudinal damage increased slowly. When the strain exceeded 2300 με, the longitudinal damage progressed quickly, ultimately leading to failure. Additionally, the simulation results indicated that no apparent damage evolved during unloading process.

Fig. 17 shows the damage state of each component of the composites at different loading stages. As shown in Fig. 17, at initial loading stage (e.g., the 3rd cycle), no damage occurred in any component of the composites. As the applied load increased and exceeded the matrix cracking stress (e.g., the 6th cycle), matrix damage initially occurred near the overlap region of the fiber bundles. Simultaneously, inelastic strain occurred. When the increase of the load (e.g., the 9th cycle), matrix damage gradually extended in the direction perpendicular to the loading direction, and the damage area merged into a large-scale region. During this stage, inelastic deformation was clearly observed at the warp yarns in the fiber bundle overlap region. However, no significant longitudinal damage in the warp yarns or transverse damage in the weft yarns was observed. As the load approaches the ultimate strength, matrix damage further evolved, and significant inelastic deformation was observed at the overlap region of fiber bundles. At the same time, extensive transverse damage occurred in the weft yarns along the direction perpendicular to the loading. As the transverse damage in the weft yarns extended to the overlap region of the fiber bundles, longitudinal damage occurred in the warp yarns, leading to the final failure of the composites. Moreover, the damage area was similar to the fracture morphology obtained by experiments. The break area was located at the overlap of fiber bundle. The similar failure modes could be obtained by experiments, as shown in Fig. 6. These simulation results confirmed that the model accurately predicted the entire progression from damage initiation to propagation, and the practical failure behavior of the composites.

5. Conclusions

This study investigated the cyclic mechanical behavior of C/SiC composites through experimental and numerical simulation approaches. Firstly, incremental loading/unloading tests were conducted using AE method and DIC, with AE signals processed via Pearson correlation analysis, optimal clustering numbers determined by CH and DB indices, and *k*-means clustering applied to identify typical damage modes. Secondly, based on the damage modes observed in experiments, a coupled anisotropic elastic-plastic damage constitutive model was proposed for fiber bundles and implemented by implicit method, while a linear elastic damage model was adopted for the brittle matrix. Finally, a mesoscale RVC finite element model was analyzed and successfully predicted mechanical behaviors of C/SiC composites including stress-strain response, residual deformation, unloading modulus, damage evolution process and failure modes. The main conclusions are summarized as follows.

(a) The mechanical response and damage process of C/SiC composites under cyclic loading were investigated. Clustering analysis of AE signals was performed to identify the typic four damage modes: matrix cracking [0–180 kHz], fiber pull-out [180–290 kHz], interface debonding [290–390 kHz], and fiber breakage [430–900 kHz].

(b) A coupled anisotropic elastic-plastic damage constitutive model was developed and implemented with implicit numerical method. Considering various damage modes observed experimentally, a coupled anisotropic elastic-plastic damage constitutive model was established. By deriving the consistent tangent stiffness matrix of this model, an implicit solution method was employed to ensure computational accuracy and efficiency.

(c) The mechanical response of C/SiC composites under incremental loading/unloading process was predicted. The mechanical behaviors of C/SiC composites under cyclic loading, including stress-strain response, residual deformation, unloading modulus, damage evolution process and failure modes were predicted. The simulation results agreed with the experiment results.

In summary, this study lays a solid foundation for analyzing the reusable capability of CMCs, but further validation should be conducted, such as testing under variable amplitude loading or temperature variations. Additionally, hysteresis behavior is a key aspect of understanding the material's nonlinear mechanical response and energy dissipation during cyclic loading. The proposed model should be improved to predict this behavior, so as to offer critical technical support for the design, performance evaluation, and engineering application of reusable CMCs.

CRediT authorship contribution statement

Qi Zhang: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Jingran Ge:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization. **Bingyao Li:** Formal analysis. **Shuwei Zhao:** Investigation. **Zengfei Liu:** Investigation. **Jun Liang:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compositesb.2026.113395.

Data availability

Data will be made available on request.

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